iPC (individualizedPaediatricCure) platform

A cloud-based architecture for the visualization, sharing, and analysis of paediatric cancer data aligned to the GA4GH standards.

GA4GH 10th PLENARY

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The iPC project aims to predict individual therapies for children with cancer. To this end, computational platform has been built, allowing users the visualization of cancer-related datasets through a data catalogue system based on Arranger, and a subsequent analysis via the Virtual Research Environment.

The prototype of the iPC data access framework is shown, which has the objective of controlling access to the private data hosted in the platform. The iPC data access framework not only aims to simplify the data access requests procedures but also allow data owners to retain control over their datasets. By design, the system enhances interoperability by leveraging the OpenIDConnect protocol and the Global Alliance for Genomics & Health Passports/Visas specification, and thereby opening the door to integrate with external systems such as the NIH (i.e. Cavatica) or EGA.

KEYCLOAK OpenID

Global Alliance for Genomics & Health — Collaborate. Innovate. Accelerate.

Virtual Research Environment CAVATICA

Passport Broker

Keycloak is an open source solution that manages authentication and authorization in the iPC platform which has been adapted to potentially act as a Passport broker. Thus, this service is able to **interact with the Passport Visa issuers** (see POSTER #) **and append Visas to a Passport**.

Passport Clearinghouse

The passport clearinghouses refers to the Passport consumers that **evaluates whether or not access should be granted**. In the iPC project, two analysis platforms are expected to play this role, the Virtual Research Environment (Europe) and Cavatica (USA).

Passport Clearinghouse

Passport

Passport Broker

Visas

Passport Visa Issuer

Assertions

Passport Visa Assertion Source

Datasets

Passport Visa Issuer

The Permissions-API service is able to **store** assertions **and dispatch Passport Visas** (Passport Visa Issuer). The Permissions-API is accessible via REST by authorized iPC users or external clients such as the Passport Brokers.


```
{
    "format": "PLAIN",
    "iss": "",
    "sub": "1201754",
    "jti": "urn:uuid:7f9bde80-...",
    "ga4gh_visa_v1": {
        "type": "ControlledAccessGrants",
        "asserted": 1647255595,
        "value": "EGAD00000000028",
        "source": "EGAC00001000514",
        "by": "dac"
    }
}
```


Data storage

The data storage is based on Nextcloud an open source solution for data storage that provides an intuitive user interface for data submissions.

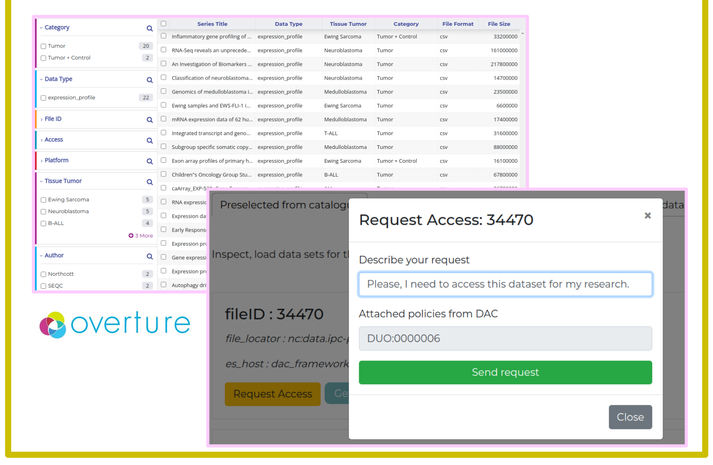
BSC EUROPEAN GENOME-PHENOME ARCHIVE

Nextcloud

Submissions metadata

Data Catalogue Portal

The Data Catalogue Portal is based on Arranger. The portal provides data visualisation tools, and also, functionalities for exposing relevant metadata into the analysis platforms and/or triggering data access requests.



Data Requests

Passport Visa Assertion Source: Data Access Committees

The Data Access Committes members can access to the Data Access Committee Portal (Figure 1) in order to **accept incoming data access requests and create assertions** (Passport Visa Assertion Source) or reject them instead, define data policies, and inspect user's dataset permissions in an interactive way. The different **Data Access Committees are created** by system administrators in the Data Access Committee Management Portal (Figure 2) upon data submissions.

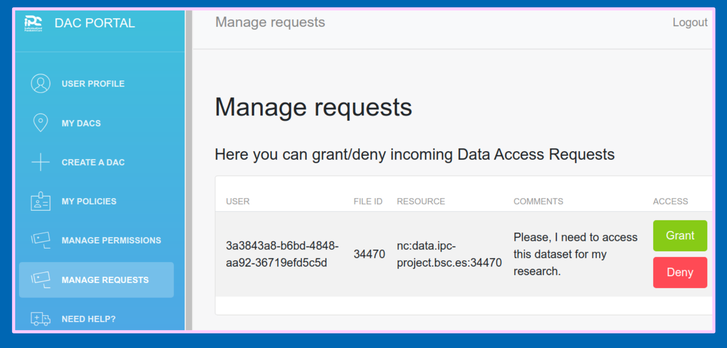


Figure 1: Data Access Committee Portal

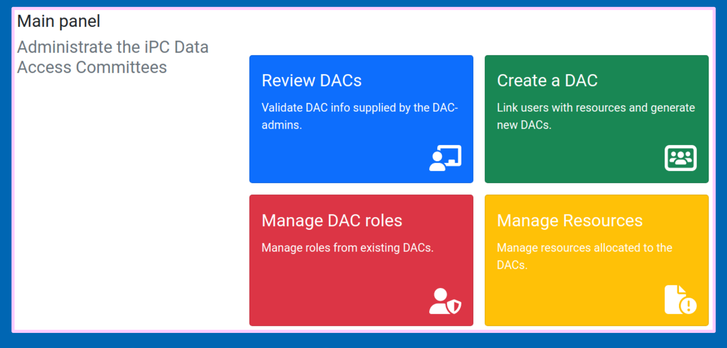


Figure 2: Data Access Committee Management Portal

INB Barcelona Supercomputing Center Centro Nacional de Supercomputación CRG Centre for Genomic Regulation XLAB

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